

Your Oracle workspace

Docker, Caddy & shared login

Deployment record: **8 October 2026** (Asia/Kolkata). This guide covers the completed dashboard and Portainer stages. Blog and learning-note editing await separate approval.

Resource	Address and state
Portfolio	https://rithvik.ddns.net/ - public, SMTP configured
Admin dashboard	https://rithvik.ddns.net/admin/ - shared login
Portainer CE 2.45.2	https://rithvik.ddns.net/portainer/ - shared login
Authelia 4.39.28	https://rithvik.ddns.net/auth/ - password sign-in
Writing tools	Stage 3 - not installed

The route from your browser

No-IP resolves the hostname to **144.24.135.201**. Caddy receives HTTPS, manages certificates, and sends requests to the correct container. Node serves the public portfolio and dashboard. Authelia verifies your identity. Portainer manages Docker on Oracle.

Only Caddy publishes web ports 80 and 443. The app, Authelia, and Portainer use private container ports. No wildcard hostname is needed for paths on the same domain.

The one command for future updates

```
bash /opt/portfolio_redeploy.sh
```

Run it **inside Oracle SSH** after committing your changes, merging them into main, and pushing. The server checkout is /opt/portfolio. A Git push alone does not trigger deployment.

Docker files and local preview

File	Use
Dockerfile	Builds the Node image, generates pages, checks links, and runs tests.
compose.yaml	Local public-site preview at http://localhost:8080 .
compose.ip.yaml	Earlier temporary HTTP preview at a public IP.
compose.production.yaml	Oracle app, Caddy, Authelia, Portainer, networks, and persistent storage.
.dockerignore	Allows only required inputs into the app image.
deploy/Caddyfile.docker	Active Docker routing and HTTPS configuration.
deploy/Caddyfile	Alternative host/systemd template; not the active Docker file.

A **Dockerfile** builds an image. **Compose** runs services together, defining their ports, environment, networks, and storage. Build creates the image; up creates or updates containers; up --build does both. Restart alone does not rebuild code.

Docker Desktop: run from the repository folder

```
docker compose config --quiet
docker compose up --build -d --wait
docker compose ps
```

Open **<http://localhost:8080>**. Port 4173 belongs to the container; host port 8080 forwards to it. Local Compose does not start the shared-login tools. The optional local .env supplies SMTP settings; without it, the contact form uses its LinkedIn fallback.

```
docker compose logs --tail=100 app
docker compose down
```

For direct Node development, use pnpm install, pnpm build, pnpm check, pnpm test, and pnpm dev. That preview uses <http://127.0.0.1:4173>. Use one Compose configuration per deployment; these files are standalone alternatives.

Redeploy from main

1. Commit local development changes, including generated public pages if you changed content or templates. **2.** Merge V2 or your development branch into main and push. **3.** Connect to Oracle and run the existing wrapper:

```
bash /opt/portfolio_redeploy.sh
```

The wrapper runs `/opt/portfolio/portfolio_redeploy.sh`. You do not need to paste a new script into SSH. The server uses main; do not repeat the earlier branch migration.

Step	What the script does
Git	Locks deployment, fetches main, and permits a fast-forward update.
Preparation	Preserves existing private files; creates missing Portainer/OIDC keys.
Validation	Checks Compose, Authelia, and Caddy before restarting services.
Build	Generates 5 pages, checks 78 links, and runs 15 app tests.
Integration	Starts auth and Portainer; applies SSO settings and verifies Docker access.
Activation	Updates the stack, reloads Caddy, and checks public/private routes.

Copied server files

Stage 1 and 2 were copied to Oracle before merging into main. Commit and merge all matching code, deployment files, docs, and the PDF. The script stashes copied files only if they exactly match origin/main; otherwise it stops and names the difference. Preserve private directories and reconcile the named file.

Server-only configuration changes

```
bash /opt/portfolio_redeploy.sh --no-pull
```

This rebuilds the current checkout without fetching. Authelia is recreated, so sign in again afterward. The app may briefly restart. Databases, keys, and certificates persist. Changing your computer's `.env` does not update Oracle's `.env.production`.

Administrator login

Open **<https://rithvik.ddns.net/admin/>**. Sign in as **rithvik** using the existing password. The private local record is Portfolio admin login.txt beside your Oracle SSH key. The server copy is .auth/bootstrap-login.txt. Keep credentials in your password manager.

Session behavior

Authelia expires sessions after 30 minutes of inactivity or eight hours total. Remember-me is disabled. Restarting auth ends its in-memory sessions. This setup uses password authentication; TOTP, WebAuthn, and email password reset are not configured.

How the dashboard is protected

Caddy removes visitor-supplied identity headers, checks Authelia, and passes the verified identity to Node. Node also requires the exact trusted Caddy IP, the configured username, and the admins group. The dashboard is marked no-store and noindex. Direct Node access does not bypass these checks.

Change or recover the shared password through SSH

```
cd /opt/portfolio
sudo docker compose --env-file .env.production \
  -f compose.production.yaml exec auth \
  authelia crypto hash generate argon2 --parallelism 1
```

Enter the new password interactively, not as a command argument. Back up .auth/secrets/users.yml privately, then edit only the account's password hash. Preserve valid JSON-compatible YAML, the username, and the admins group.

```
sudo docker compose --env-file .env.production \
  -f compose.production.yaml restart auth
```

Verify the new login and update your password manager. Bootstrap records are not updated automatically. Keep .auth/secrets/storage-key with its matching database; replacing that key alone prevents decryption.

Portainer and shared sign-in

Select **Open Portainer** on the dashboard, or visit <https://rithvik.ddns.net/portainer/>. If Portainer presents a login page, select **Login with OAuth**. Your existing Authelia session is reused; you do not enter another password. Open the connected **Oracle Docker** environment.

What you can manage

The rithvik account is a Portainer administrator. It can view containers and logs, open consoles, and manage images, volumes, and networks. The mounted Docker socket provides control over Oracle containers. Keep lasting changes to the portfolio stack in this repository; Compose can replace manual container settings on redeployment.

Two related sessions

Portainer creates its own eight-hour session. Caddy checks the Authelia session on each new request, so a Portainer token alone cannot bypass shared login. Portainer logout clears its session and redirects to Authelia logout. Dashboard sign-out blocks subsequent private requests but does not itself revoke the Portainer token. Close open console WebSockets when signing out.

Integration details

Portainer runs with `--base-url=/portainer`. Caddy authorizes that path before stripping its prefix. Translation requests under `/locales/` redirect through the protected Portainer path. The OIDC client is restricted to admins, maps `preferred_username` to rithvik, and disables automatic account creation. Its callback includes the trailing slash.

Recovery account

The separate initial account is **admin**. Its generated password is in `.portainer/recovery-password` and the private local Portainer recovery `login.txt`. The redeploy helper uses it on the unpublished API. Ordinary access uses rithvik with OAuth. If you change the recovery account password, update both private records to match; the startup file does not reset an existing database password.

Detailed recovery and SSH tunnel instructions are in **deploy/PORTAINER.md**. No Portainer host ports are published. Its data survives rebuilds in `.portainer/data/`.

Configure Caddy for new sites

Edit **deploy/Caddyfile.docker**. Add the application service to Compose and a deliberate network shared with Caddy. Proxy to the service name and internal port, such as tool:3000. Leave the application port unpublished.

A public site on a new hostname

Create a DNS record pointing tool.example.com to Oracle, then add a top-level block after replacing the example values:

```
tool.example.com {  
    encode zstd gzip  
    reverse_proxy tool:3000  
}
```

A public application under the existing hostname

If the app supports a base path, add this inside the existing hostname block before the fallback handler. No extra DNS record is needed:

```
redir /tool /tool/ 308  
handle_path /tool/* {  
    reverse_proxy tool:3000  
}
```

handle preserves the path; **handle_path** strips its matched prefix. Apps must generate assets and redirects for the public base path. Authelia preserves /auth; Portainer needs /portainer stripped. The examples above are public and do not add login protection.

For private tools

Add an Authelia access rule and forward_auth before the application handler. When stripping a prefix, use an explicit route so authentication checks the original path first. Also configure the application's supported OAuth/OIDC integration if it has its own login. A dashboard link or proxy gate alone does not establish application SSO.

Validate the file and reload Caddy only after validation succeeds. For changes to Compose, auth, or DOMAIN, use the redeploy script. Blog and learning-note services are still a separately approved next stage.

Settings, storage, and domains

Location	Preserve or update
.env.production	Private domain, SMTP, proxy network, and account settings on Oracle.
.auth/secrets/	User hash, session key, storage key, OIDC HMAC/RSA key and client hash.
.auth/data/	Persistent SQLite auth state and filesystem notification output.
.portainer/	Recovery password, OAuth client credentials, and persistent data/.
portfolio_caddy_data	Certificate and ACME account state in a named Docker volume.
portfolio_caddy_config	Caddy runtime state in a named Docker volume.

Private parent directories use mode 700; credential files use mode 600. Portainer runs as root for socket access and owns its database files. Backups need sudo. Secrets stay out of Git, the app image, and public file serving. Use Compose config --quiet; full resolved configuration may reveal SMTP credentials.

Move to a purchased domain

Point the new hostname to Oracle. Set DOMAIN in .env.production to the hostname only, with no scheme or path. Keep persistent storage, then run:

```
bash /opt/portfolio_redeploy.sh --no-pull
```

Compose derives the public origin, build URL, auth URLs, and Portainer trusted origin. The helper updates Portainer's authorization, token, user-info, callback, and logout URLs. Caddy obtains the hostname's certificate. Sign in again and check both private applications.

Why the domain remains in configuration

DNS finds an IP address. HTTPS still carries the hostname for TLS and HTTP routing. Caddy selects the certificate and site; apps use the public origin for redirects, callbacks, and cookies. This deployment redirects HTTP on the raw IP to the domain; it is not configured for raw-IP HTTPS.

Keep the No-IP hostname renewed. Update DNS if Oracle's public IP changes. No Cloudflare tunnel, Pages deployment, Worker, or dynamic DNS updater is installed here.

Backups and recovery

Back up `.env.production`, the complete `.auth/` and `.portainer/` directories, the matching source/image version, and both Caddy volumes. Store backups privately outside Git. Do not use `down --volumes` as a backup command.

Consistent private-state backup inside Oracle SSH

```
cd /opt/portfolio
umask 077
backup=$(mktemp -d \
  /home/opcl/.config/portfolio-deploy/private-state-XXXXXXX)
dc() { sudo docker compose --env-file .env.production \
  -f compose.production.yaml "$@"; }
dc stop auth portainer
sudo tar -czf "$backup/private-state.tar.gz" \
  .env.production .auth .portainer
sudo chmod 600 "$backup/private-state.tar.gz"
dc start auth portainer
dc ps
```

Start both services again if archiving fails. The public portfolio stays available while private tools pause. This command does not archive Caddy volumes; preserve them separately for a full migration.

Restore as a matched set

Stop the relevant services before restoring their databases. Restore corresponding keys, passwords, client credentials, data, owners, and permissions together. Use the compatible source and pinned image. Preserve current state before rollback. An older app image alone does not undo proxy or auth changes.

Deployment recovery points

The stage 2 directory is:

**/home/opcl/.config/portfolio-deploy/
portainer-stage2-6erYb3rT/**

It holds the earlier source/private configuration, a SQLite auth backup, and the post-deployment private-state backup. The previous app image is `rithvik-portfolio:before-portainer-stage2`.

The earlier stage 1 backup remains in `admin-stage1-YzXVP4Gu` under the same parent directory. These are manual deployment recovery points, not scheduled backups. The Markdown guides list the exact files and procedures.

Checks, first setup, and references

Routine checks inside SSH

```
cd /opt/portfolio
sudo docker compose --env-file .env.production \
  -f compose.production.yaml ps
curl -I https://rithvik.ddns.net/
curl -I https://rithvik.ddns.net/admin/
curl -I https://rithvik.ddns.net/portainer/
curl https://rithvik.ddns.net/api/contact
```

Expect public 200 responses and private redirects when signed out. Contact available:true confirms configured delivery, not inbox receipt. Inspect service logs for 502 errors, auth failures, or socket problems. No test email was sent during deployment.

Problem	Next check
SSH key missing	Use the full quoted key path on the current computer.
Public timeout	DNS, Caddy, Oracle ingress, and host firewall for TCP 80/443.
Blank Portainer page	Trailing slash, base URL, and Caddy prefix handling.
OAuth failure	Matching client secret/hash and exact callback URL.
Redeploy refuses Git changes	Merge matching server copies into main or reconcile the named file.

First setup on another Linux server

Install Docker/Compose, Git, Python 3, OpenSSL, curl, and flock. Clone main, configure DNS/firewall, and privately create .env.production from deploy/production.env.example. Set the correct UID/GID and account. Run scripts/setup-admin.py once, then portfolio_redeploy.sh --no-pull as the deployment user. The current Oracle server is already initialized.

Verified for stage 2

All 15 app tests passed. Production build, config and link checks passed. One shared login completed Portainer OAuth as an administrator and listed all four Oracle containers. Repeated setup preserved accounts and keys. Signed-out and forged-header requests were blocked; shared logout protected both apps. Public/contact health remained available.

Further reading

Local: README.md, deploy/DOCKER.md, deploy/ADMIN.md, deploy/PORTAINER.md.

Official: Portainer OAuth settings; Authelia Portainer integration; Caddy path routing.

Outage recovery and memory

The original nine-page guide remains applicable. This addendum records the host-level recovery change made after the same-day outage.

Verified after the reboot

The portfolio and authentication health endpoint returned HTTP 200. Dashboard and Portainer requests correctly redirected signed-out visitors. All four containers restarted, and the complete shared-login, OAuth, container-listing, and logout checks passed. CPU load was about 0.1, available RAM about 415 MB, and root-disk use 36%.

What the saved logs showed

The host ran out of memory three times: 7 October at 23:31 UTC, and 8 October at 06:56 and 09:29 UTC. Each event terminated the dnf process inside dnf-makecache.service. At 09:29, DNF used roughly 596 MiB resident RAM, while only 252 KiB of the host's approximately 2.5 GiB swap remained free.

The last event overlaps the outage and strongly suggests memory and swap exhaustion. Earlier events preceded Portainer installation. SSH was unavailable during the failure, so no live resource sample could be taken. The previous boot journal was lost, but /var/log/messages retained these kernel records.

Targeted change applied

The recurring package-metadata cache job was stopped and its timer disabled:

```
sudo systemctl disable --now dnf-makecache.timer
sudo systemctl stop dnf-makecache.service
```

Both are now inactive, and the timer is disabled across reboots. Package installation and upgrades remain available. No repositories or update-agent settings were changed. Manual package operations can still require substantial memory; plan them for a maintenance window on this 1 GB VM.

Recovery and scope

The previous unit definitions and state are saved under:

**/home/opcl.config/portfolio-deploy/
dnf-cache-outage-MGfiPW0L/**

After addressing memory capacity, scheduled caching can be restored with:

```
sudo systemctl enable --now dnf-makecache.timer
```

This is an Oracle host setting, separate from Compose. The redeploy script does not apply it to new servers. The post-reboot check did not reboot the VM or restart its containers. Full incident details are in deploy/DOCKER.md.

Reference: DNF makecache command.